Senior Three Biology Study Notes

Selected revision topics • Masaka District Digital Learning Hub

1. Cell structure

• The cell is the basic structural and functional unit of living organisms.

• Common structures include the cell membrane, cytoplasm and nucleus. Plant cells also commonly have a cell wall, large permanent vacuole and chloroplasts in photosynthetic tissues.

• The cell membrane controls movement of substances into and out of the cell.

2. Nutrition in plants

• Photosynthesis is the process by which green plants manufacture carbohydrates using light energy.

• The raw materials are carbon dioxide and water. Light and chlorophyll are required.

• A simplified word equation is: carbon dioxide + water → glucose + oxygen.

3. Nutrition in humans

• A balanced diet contains appropriate amounts of carbohydrates, proteins, fats, vitamins, minerals, fibre and water.

• Digestion breaks large food molecules into smaller soluble molecules that can be absorbed.

4. Ecology

• An ecosystem includes organisms and the physical environment with which they interact.

• A food chain shows the transfer of energy as one organism feeds on another.

5. Check your understanding

• Name three structures found in a typical animal cell.

• State two requirements for photosynthesis.

• What is the role of the small intestine in digestion?

• Define an ecosystem.

Learning note: Prepared as a general revision aid. Confirm topic coverage with your teacher and current school scheme of work.